

Network Security Assessment

Modular Assignment 4

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IHC - Practical Approach to Cyber Security (PACS)

1. Introduction and Objective

This assessment focuses on the detection and mitigation of common network security threats using standard network-analysis tools and controlled, authorized lab techniques.

The assignment scope follows the brief: establish a network threat model, demonstrate ping, Nmap, traceroute, Wireshark and tcpdump, analyze protocol-based attacks including blind spoofing and TCP SYN flooding, and examine DDoS, botnet and crimeware risks with firewall and VPN defenses.

No uncontrolled attack traffic was generated against third-party systems. Evidence in this package is either command output from a local/authorized lab or clearly labeled simulated security telemetry.

2. Threat Model and Scope

Assumed environment: a small enterprise network with Internet connectivity, a perimeter firewall, internal hosts, an application server, remote users connecting through a VPN, and monitoring sensors.

Threat actors considered include external attackers performing reconnaissance, distributed botnets generating availability attacks, compromised internal hosts, and unauthorized remote users.

Primary security objectives are confidentiality of administrative traffic, integrity of network communications and configurations, availability of services, traceability through logs, and controlled remote access.

3. Common Network Threats

Reconnaissance: ICMP discovery, route discovery and TCP/UDP port scanning can reveal reachable systems and exposed services.

Protocol attacks: TCP SYN flooding consumes resources through large numbers of incomplete connection attempts. Blind spoofing uses forged source information so that responses are sent to an address other than the attacker.

DDoS: distributed sources can overwhelm bandwidth, connection tables, CPU, or application capacity. Botnets can provide the distributed infrastructure for such activity.

Crimeware: malware can establish command-and-control communication, steal credentials, proxy traffic, or recruit hosts into a botnet.

4. Network Analysis Tool Demonstration

ping verifies basic IP reachability and round-trip response behavior.

Nmap identifies reachable ports and services on an explicitly authorized target. The package uses a localhost/controlled-target example.

traceroute/tracert displays the sequence of routing hops toward an authorized destination and helps identify path changes.

Wireshark provides packet-level inspection. Useful filters include `tcp.flags.syn == 1 && tcp.flags.ack == 0` for SYN observations, `icmp` for ICMP traffic, and `dns` for DNS traffic.

tcpdump provides lightweight command-line capture and filtering. Example filters focus on TCP SYN and ICMP traffic for a controlled lab.

The tools complement each other: ping checks reachability, Nmap maps exposed services, traceroute examines the path, while Wireshark/tcpdump provide packet-level evidence.

5. Protocol Attack Analysis

TCP SYN flood: a TCP connection begins with SYN, followed by SYN/ACK and then ACK. A flood of SYN packets that do not complete the handshake can consume connection-state resources. Indicators include a high SYN-to-completed-connection ratio, many half-open sessions, source diversity, and a sharp traffic-rate increase.

Blind spoofing: an attacker forges source information without seeing the returning traffic. Detection uses source-address validation, asymmetric or impossible routing observations, unexpected traffic patterns, and ingress/egress filtering.

ICMP abuse: excessive echo requests can consume bandwidth and processing resources. Detection uses rate, source distribution, packet-size patterns, and correlation with other traffic anomalies.

6. DDoS, Botnet and Crimeware Analysis

DDoS attacks may originate from many compromised systems. Indicators include abrupt volume changes, many source addresses, repeated destination targeting, unusual distribution, and service-level degradation.

Botnet activity can be detected by repeated outbound connections to suspicious destinations, beacon-like periodicity, unusual DNS lookups, unexpected destination ports, and hosts communicating with command-and-control infrastructure.

Crimeware creates additional risk because a compromised endpoint can become both a source of malicious traffic and a pivot into internal systems. Endpoint isolation, egress controls, DNS monitoring, IDS/IPS, patching and least privilege reduce this risk.

7. Defense Mechanisms

Firewall: apply least-privilege rules, restrict unnecessary inbound services, rate-limit abusive traffic, and log denied/threshold events.

IDS/IPS: use Suricata or equivalent rules to identify high-rate SYN/ICMP patterns and generate alerts. Thresholds should be tuned to normal traffic baselines.

VPN: require authenticated remote access, strong cryptographic settings, restricted routes, current software, and centralized logging.

Network architecture: segment sensitive systems, restrict management access, place public services behind appropriate controls, and monitor trust boundaries.

Operational controls: maintain rule reviews, patch systems, monitor alerts, train users, retain logs according to policy, and test incident-response procedures.

8. Evidence and Observations

The evidence folder contains representative outputs and logs for the required tools. Because the assignment permits simulated or real datasets, simulated telemetry is explicitly marked where used.

Example SYN-flood telemetry shows repeated SYN requests without corresponding completed handshakes. Example reconnaissance output demonstrates how exposed services can be interpreted. Example IDS-style alerts show how a detection control can translate traffic patterns into actionable events.

The evidence is designed for reproducible academic demonstration rather than a claim that a production network was attacked.

9. Risk and Mitigation Summary

Reconnaissance — exposure of services and topology; controls: firewall minimization, segmentation, monitored scanning and hardened services.

TCP SYN flood — resource exhaustion and availability loss; controls: rate limiting, SYN protection, upstream DDoS protection and IDS/IPS alerting.

Spoofing — bypass of source-based trust and reflection abuse; controls: ingress/egress filtering and anti-spoofing ACLs.

DDoS/botnet — bandwidth and service exhaustion plus compromised hosts; controls: upstream filtering, rate limiting, IDS/IPS, endpoint isolation and egress monitoring.

Unauthorized VPN access — remote compromise; controls: strong authentication, restricted access, secure cryptography, patching and audit logging.

10. Conclusion

The assessment demonstrates an end-to-end network-security workflow: model threats, collect traffic evidence, interpret protocol behavior, identify attack indicators, and apply layered controls. The strongest practical approach is coordinated visibility, preventive controls, detection and response.

The package maps directly to the assignment deliverables and keeps demonstrations within controlled or simulated environments.

References and Resources

TCS IHC - Practical Approach to Cyber Security, Modular Assignment 4: Network Security (provided assignment brief).

DDoS Attack Detection Dataset (Kaggle) — listed in the assignment brief.

Wireshark Sample Captures — listed in the assignment brief.

Nmap Documentation and Reference — listed in the assignment brief.

Digital Attack Map — listed in the assignment brief.

Threat Model Matrix

Threat	Vector	Indicator	Controls
Reconnaissance	ICMP/TCP/UDP probes	Repeated probes/scans	Segmentation, firewall minimization, monitoring
SYN flood	TCP listener	High SYN rate, incomplete handshakes	Rate limiting, SYN protection, IDS/IPS
Blind spoofing	Forged IP source	Invalid source patterns	Ingress/egress filtering, anti-spoofing
DDoS	Bandwidth/service capacity	Traffic spike, many sources	Filtering, rate limiting, DDoS protection
Botnet/C2	Compromised endpoint	Periodic outbound connections	EDR, DNS monitoring, egress controls
VPN compromise	Remote access	Failed logins/unusual source	Strong authentication, restricted routes, logging

Network Tool Command Reference

Tool	Controlled example
Ping	ping 127.0.0.1
Nmap	nmap -sT -sV 127.0.0.1
Traceroute Windows	tracert 127.0.0.1
Traceroute Linux	traceroute 127.0.0.1
tcpdump	sudo tcpdump -ni any 'tcp[tcpflags] & tcp-syn != 0'
Wireshark SYN	tcp.flags.syn == 1 && tcp.flags.ack == 0
Wireshark ICMP	icmp